

EDUCATION	Northeastern University	Boston, USA
	<i>Ph.D. in Marine and Environmental Sciences</i>	2024 – Present
	• Advisor: Prof. Cristina Schultz • Dissertation: Linking Terrestrial Inputs, Coastal Benthic Fluxes, and Carbon Dynamics: A Multi-Scale Study from the Plum Island Estuary to the Gulf of Maine	
	China University of Geosciences (Beijing)	Beijing, China
	<i>M.E. in Geological Engineering</i>	2018 – 2021
	Zhejiang University	Hangzhou, China
	<i>B.E. in Environmental Engineering</i>	2013 – 2017
POSITIONS	Chinese Academy of Sciences	Nanjing, China
	<i>Research Associate, Nanjing Institute of Geography and Limnology</i>	2021 – 2024
	China University of Geosciences (Beijing)	Beijing, China
	<i>Lab Manager, MC-ICP-MS Laboratory</i>	2019 – 2020
PUBLICATIONS	[1] Wu, Y., Chen, S., Ye, Q. , Jiang, L., Chang, Y., Huang, W., Liu, C., Ma, S., Jiang R., Fei, D., Xu, H., Shi, X., and Chen, K., 2024. Coverage and diversity of submerged macrophytes drives sediment bacterial community in Dongshan Bay, Lake Taihu. (2025.02.27: Major revision from <i>Freshwater Biology</i>)	
	[2] Wu, Y., Ye, Q. , Chen, S., Chen, K., Ma, S., Chang, Y. and Shi, X., 2024. Response of phytoplankton functional group to spring drought in a large subtropical reservoir. <i>Journal of Oceanology and Limnology</i> , pp.1-11.	
	[3] Wu, Y., Ye, Q. , Chen, S., Jiang, L., Chen, K., Ma, S., Chang, Y., Liu, J. and Shi, X., 2024. Climate changes drive phytoplankton community through complementarity and selection effects in a large mesotrophic reservoir. <i>Journal of Oceanology and Limnology</i> , pp.1-15.	
	[4] Chen, S., Jiang, L., Ma, S., Wu, Y., Ye, Q. , Chang, Y., Ye, Y. and Chen, K., 2024. Response of a submerged macrophyte (<i>Vallisneria natans</i>) to water depth gradients and sediment nutrient concentrations. <i>Science of The Total Environment</i> , 912, p.169154.	
	[5] Yu, C., Wang, H., Blaustein, R.A., Guo, L., Ye, Q. , Fu, Y., Fan, J., Su, X., Hartmann, E.M. and Shen, C., 2022. Pangenomic and functional investigations for dormancy and biodegradation features of an organic pollutant-degrading bacterium <i>Rhodococcus biphenylivorans</i> TG9. <i>Science of The Total Environment</i> , 809, p.151141.	

Northeastern University

2024 – Present

Department of Marine and Environmental Sciences

As a Ph.D. student working with Prof. Cristina Schultz, I'm dedicated to evaluating the influence of benthic processes and terrestrial input on coastal nutrient cycles and water quality. In particular, I am:

- ◆ Compiling and integrating data from long-term observational records, hydrological datasets, and high-resolution ocean biogeochemistry models to build a comprehensive database for analyzing nutrient fluxes, oxygen dynamics, and sediment properties.
- ◆ Investigating the relationships among key environmental factors in PIE, particularly the role of benthic processes in regulating nutrient cycling and oxygen availability.
- ◆ Comparing patterns across estuaries by examining how PIE's nutrient fluxes, oxygen dynamics, and sediment properties differ from other northeastern U.S. estuaries, while exploring connectivity between these coastal systems.
- ◆ Assessing the impact of extreme events (e.g., storms, heatwaves) on estuarine dynamics and developing guidelines for future long-term monitoring and data integration to improve our understanding of estuarine resilience.

Chinese Academy of Sciences

2021 – 2024

Nanjing Institute of Geography and Limnology

As a research associate working with Prof. Kaining Chen and Prof. Xiaoli Shi, I focused on exploring the relationships between phytoplankton and the water column in lakes and reservoirs. In these projects, I:

- ◆ Investigated the mechanisms behind algal blooms caused by changes in nutrient concentrations, meteorological and hydrological conditions using long-term monitoring data series.
- ◆ Studied the impact of global warming on drinking water sources.

China University of Geosciences (Beijing)

2018 – 2021

Institute of Earth Sciences

The title of my master's thesis is An Improved Method for Precise Selenium Isotope Determination. Under the guidance of Prof. Jianming Zhu, I:

- ◆ Improved the testing method for precise selenium isotope determination.
- ◆ Tested and reported the selenium isotope ratios of a series of standard samples.

Zhejiang University

2016 – 2017

College of Environmental and Resource Sciences

The title of my undergrad's thesis is Hormesis Response of Polychlorinated Biphenyls to *Escherichia Coli*. Under the guidance of Prof. Chaofeng Shen, I:

- ◆ Established a microplate based, high throughput test method for *E. coli* growth.
- ◆ Identified the types of PCBs that can show the hormesis response on *E. coli* growth and investigated the mechanisms.

INTERESTS

- ◆ Relationships between climate change, human activities, and aquatic system.
- ◆ Environmental complex systems modeling using **R** and **Python**.
- ◆ Large-scale dataset modeling and simulation.
- ◆ Environmental sustainability and environmental geochemistry.

TEACHING	Northeastern University <i>Teaching Assistant, Department of Marine and Environmental Sciences</i> ◆ Served as the instructor of 2 sections of Lab for Biostatistics (ENVR 2501), an undergraduate course, each with 19 students. ◆ Delivered a one-hour lecture per section on advanced statistical methods in R (Bayesian/frequentist approaches, linear/generalized linear models, mixed-effects models, multivariate statistics, and ensemble modeling), followed by a two-hour Q&A session. ◆ Provided hands-on guidance, working closely with students on real-world data analysis to ensure they developed strong practical skills in R.	Spring 2025
ACADEMIC SERVICES	Freshwater Biology <i>Peer Reviewer</i> ◆ Evaluated and provided feedback on manuscripts related to freshwater ecosystems, assessing scientific rigor, methodology, and ecological significance.	2024
SCHOLARSHIPS	◆ Full Scholarship Ph.D. Program, Northeastern University. ◆ Second-Class Academic Scholarship, China University of Geosciences.	2024 2019, 2020
SKILLS	Programming & Software ◆ Proficient in R and Python for data analysis and computational tasks. ◆ Experienced in large-scale modeling in high-performing computing environment. ◆ Skilled with ArcGIS, MIKE, Origin, L^AT_EX, Microsoft Office, Adobe Creative Suite. Laboratory & Field work ◆ Proficient in field navigation for locating sampling sites using portable GPS devices. ◆ Proficient in sample collection and preservation (water, sediment, phytoplankton, zooplankton, aquatic plants, etc.). ◆ Proficient in chemical laboratory skills, including sample preparation and analysis. ◆ Experienced in ultra-clean laboratory procedures such as isotope sample purification. ◆ Skilled in the operation and maintenance of experimental instrumentation including UV-Spectrophotometer, Atomic Fluorescence Spectrometer (AFS), Inductively Coupled Plasma Mass Spectrometer (ICP-MS), and Multi-Collector ICP-MS (MC-ICP-MS). Languages ◆ Chinese: Native. ◆ English: Full professional proficiency.	
ACTIVITIES	Northeastern University <i>Member of the Ph.D. Student Council, College of Science</i> ◆ Serve as a liaison between the Dean's Office and Ph.D. students, ensuring their concerns and suggestions were represented. ◆ Build community among Ph.D. students across different departments through events and programs, fostering collaboration. ◆ Provide a platform for sharing ideas, addressing issues, and encouraging constructive dialogue within the College.	2024 - Present